Falsification-First Decipherment: A Decontaminated Inference Framework for Testing Undeciphered-Script Claims, with Linear A as a Worked Null

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Abstract

Decipherment research has a false-positive problem and no shared instrument: a small corpus, an unknown language, and many candidate readings let an analyst "translate" almost anything. The flagship survey lists fifteen challenges of script and data; none addresses the orthogonal inference-control layer — overfitting, multiple testing, contamination — a candidate sixteenth challenge. We build a falsification-first framework for it: a fabricated-language negative-control family, multiple-comparison deflation, a literature index separating published from not-indexed signs, and pre-registration with a mechanically-graded gate. This is a methods paper; we assert no decipherment. The machinery calibrates: on a tested null the gate's false-graduation rate is 0.6% (3/500); the identical morphology test finding no power on Linear A recovers Mycenaean (Linear B) inflection (9/16 affixes above the bigram floor), supporting the interpretation that Linear A's short forms contribute materially to its lack of power. Across probes it returns differentiated, calibrated outcomes — null, no-power, supported-structural, circular, and contamination-sensitive. Results are consistent with identifiability, rather than corpus size, being the more fundamental present constraint. The June-2026 *301 "cracked" claim is a qualitative illustration of the failure modes the gate is designed to detect.

1 Introduction

Decipherment has a false-positive problem, and the field has no agreed instrument for measuring it. The history of undeciphered-script research is, in large part, a history of uncontrolled model selection: a tiny corpus, an unknown language, and a large space of candidate readings let a determined analyst "translate" almost anything, with internal consistency on the cherry-picked set standing in for confirmation. The mechanism is not any single bad match but uncorrected multiplicity: with enough candidate roots and a small enough corpus, a pile of individually-plausible matches is the *expected* outcome under pure chance, so each individual comparison looks suggestive while the conclusion is nonsense. A standard demonstration of spurious comparison: one can "prove" *English is a Semitic language* by matching a handful of English words to Proto-Semitic roots while quietly ignoring the far larger number that do not match. The same move, applied to a real undeciphered script, has produced a long line of decipherment claims that no one can either confirm or refute.

That the discipline lacks a shared standard here is not our editorializing but the field's own assessment. The flagship computational-decipherment survey — Braović et al.'s (2024) systematic review of Bronze Age Aegean and Cypriot scripts — enumerates fifteen "major challenges" of script, language, and data (§2.1) (Braović et al., 2024). Those fifteen are primarily properties of the *object and its data* — the script, the language, the surviving corpus, and the resource/comparison-data conditions. What governs whether a *claimed reading* is believable is a different kind of thing: a property of the inference methodology and its epistemics — overfitting, multiple testing, false-positive control, and contamination. We therefore frame this axis, once and modestly, as a candidate **sixteenth challenge** — a deliberate rhetorical foil to the survey's fifteen, not a claim to a new taxonomic slot or to sole discovery. It is *under-instrumented* rather than unrecognised (§2.4): our contribution is to *consolidate and operationalize* that axis for decipherment, not to invent it. It is the methodological axis the object-level taxonomy was never meant to cover, and it is distinct from the survey's own acknowledged evaluation gap (§2.1) — an *evaluation-target* problem, separable from the false-positive / multiplicity problem; it is the latter this paper instruments.

Contribution. We build the machinery for that axis: a falsification-first, decontaminated inference framework for testing undeciphered-script decipherment claims (Braović et al., 2024). The harness applies fail-closed validation to every positive and enforces four concrete defences: a fabricated-language canary that supplies a **generator-conditional negative-control distribution** (a false-positive floor conditional on the fabricated-language generator — here a first-order sign-bigram Markov model; a broader control ensemble spanning unigram, root-template, and real-language generators is disclosed as future work, not built here); multiple-comparison deflation for the number of signs, roots, and families a search was free to try; a literature index that partitions published (`L_indexed`) from literature-unseen (`L_not_indexed`) signs so that reproducing a known value cannot masquerade as discovery; and pre-registration with a held-out gate whose verdict is computed mechanically, never by the model that proposed the reading. The evaluation gap gets a concrete, if power-limited, answer: the recurring libation *formula* — roughly 50 inscribed objects spread across peak sanctuaries and caves at more than five findspots — supports a natural leave-one-site-out cross-validation graded from held-out data — explicitly a low-statistical-power test, reported as such rather than as decisive (§2.1; Appendix A.3) (Braović et al., 2024).

This is a methods paper, and Linear A is its worked null. We do not claim a decipherment. The framework is designed to withhold validation when a proposed reading does not clear pre-registered negative controls, multiplicity adjustment, contamination checks, and held-out evaluation; when nothing survives, the calibrated null — what the corpus can and cannot support — is itself the deliverable. The working conclusion on current data is that the results are **consistent with identifiability** being the more fundamental present constraint — not a demonstrated corpus-size threshold: there is no known related language to map onto, the multiple-testing burden of candidate readings goes uncorrected, and the sign inventory is mixed and partly lossy. We do not claim additional data would be irrelevant — a null can arise from identifiability, low power, or corpus size alike. A lossy, non-injective sign-to-sound channel need not admit a decipherment at any corpus size, so we make no claim that "there is enough text to pin a reading"; the rigorous demonstration of *that* constraint is the result the field can use. The June-2026 AI-assisted "Linear A cracked" claim — reading the *301-anchored libation verb as an extinct Semitic form — serves throughout as a qualitative illustration of the failure modes the gate is designed to detect, analysed from its public, archived presentation as a claim, not a person, and neither reproducible nor held-out validated. The artifact is open and archived (§9.7).

Roadmap. Section 2 situates logos against the Braović et al. taxonomy and the computational-decipherment lineage, including the cognate-ID baseline. Section 3 specifies the harness — canary, deflation, literature index, and the pre-registration/held-out gate. Section 4 develops the information-floor diagnostic and states its identifiability limits. Section 5 works the June-2026 *301 claim as a case study of the failure modes. Sections 6–8 report the pre-registered nulls: morphology (§6); metrology, phonology, and cross-script signals (§7); and contamination and sufficiency probes (§8). Section 9 draws the discussion, limitations, and conclusion.

2 Related work

2.1 Position relative to the survey taxonomy

The reference frame for computational work on the Bronze Age Aegean scripts is Braović et al. (2024, *Computational Linguistics* 50(2):725–779), the field's first survey focused specifically on Aegean and Cypriot scripts; its only comparable predecessor is Ferrara and Tamburini (2022), and the broader machine-learning-for-ancient-languages landscape is surveyed by Sommerschield et al. (2023, *Computational Linguistics* 49(3)). Braović et al.'s Section 4 enumerates fifteen "major challenges" — from unknown writing system, reading direction and language, through small dataset, allographs and incomplete vocabulary, to unavailable parallel data and hardware — **none** of which concerns overfitting, multiple testing, false-positive control, contamination, or distinguishing a *fitted* decipherment from a *real* one. The same gap runs through Ferrara and Tamburini's (2022, *Lingue e Linguaggio* XXI.2:239–259) §3.5, which reviews the cognate-matching lineage logos builds on — Snyder et al. (2010) on Ugaritic↔Hebrew (the 29/30-sign benchmark our canary calibrates against), Berg-Kirkpatrick and Klein (2011), and Luo et al. (2019, 2021) — and frames Linear A, via Gelb and Whiting's typology (Gelb and Whiting, 1975), as a Type III script (unknown script *and* language) whose "necessary validation" is "altogether impossible," yet nowhere treats how a match is to be *deflated* for the multiplicity it was selected from. Two surveys, the same unfilled axis. We do not score this as an oversight: their challenges are properties of the object and its data, whereas the axis logos foregrounds is a property of inference *methodology* — named-but-unfilled rather than absent. logos supplies the machinery to police it (§3).

Braović et al. do gesture at a related but separable difficulty: evaluation. Because computational analyses of an undeciphered script have "nothing to compare them against," they conclude (Sec. 6.3) that "their evaluation remains an open problem." That is a distinct problem from overfitting and multiplicity control, and logos addresses it separately. Its held-out design is a concrete response — the libation formula recurs across a small set of peak-sanctuary and cave findspots, permitting a leave-one-site-out cross-validation adjudicated mechanically by `verdict.py` — with the low-power caveat flagged at the outset (§1). We locate our components in the survey's own method taxonomy (Sec. 7.1) and, where prior art exists, defer to it rather than claim novelty.

2.2 The current philological synthesis

Salgarella (2025), *Writing in Bronze Age Crete: "Minoan" Linear A* (Cambridge Elements), by the author of the SigLA database, is the most current authoritative one-author overview. It concludes that Minoan is a language **isolate** — it "does not belong in any known language family" (the families used for comparison being Indo-European, Semitic and Afro-Asiatic) — typologically agglutinative with a provisional Verb–Subject–Object order, and it records that "no bilingual text has so far been found" in a corpus of roughly 2,500 inscriptions. Crucially, Salgarella ranks the etymological/comparative method below combinatorial-contextual analysis and names **digital/statistical analysis as "the next desideratum in the field."** logos positions itself as exactly that desideratum — but under discipline, treating the isolate verdict as a central *identifiability* constraint rather than a target to be argued away.

2.3 Prior computational attempts and the search trap

The seminal automatic-decipherment result is Luo et al. (2019), which recovers a character mapping from a lost script to a *known related* language via neural minimum-cost flow. It auto-deciphered Ugaritic→Hebrew (+5.5% over prior state of the art) and translated 67.3% of Linear B→Greek cognates, but did not address Linear A and would fail there by construction: with no known cognate plaintext, the objective has nothing to optimise toward. This cognate-language precondition is the empirical form of our identifiability argument. The cognate-ID baseline logos actually runs is *not* a reimplementation of Luo's neural flow: it is a published matcher, Tamburini's (2025) CSA_OptMatcher (coupled simulated annealing), which reports the same Luo et al. (2019) accuracy metric. We adopt it as the non-neural baseline against which any Linear A cognate claim must clear.

Against that disciplined baseline stand the in-domain instances of the search trap: the dictionary-fishing programmes of Eu et al. (2019) and Loh and Perono Cacciafoco (2020), the latter explicitly a "brute force attack" on dictionaries. These are the concrete Linear-A cases the multiplicity correction is built to deflate. On the visual side, cross-script comparison is *not* our novelty: Daggumati and Revesz (2019, 2023) already applied CNN+SVM cross-script similarity, and Corazza et al. (2022) ("Sign2Vec") clustered signs unsupervised; our defensible contribution is only the controlled image-versus-sequence *asymmetry* under one harness.

2.4 The discipline lineage

Two older strands supply the honesty machinery. Packard (1974), *Minoan Linear A*, constructed nine deliberately fictitious decipherments as a null control — and the genuine Ventris-value transfer beat them only about 2:1 — anticipating our permutation-null canary. The statistical lineage is selective inference and multiple-comparison correction (Bailey and López de Prado, 2014), the correction for the number of hypotheses $\times$ signs $\times$ roots $\times$ families tried that turns a raw match-rate into a defensible one; its transfer from the finance/backtest setting to script decipherment is specified in Section 3. logos's empty cell is the intersection none of these occupy: a falsification-first, decontaminated inference framework that couples a published cognate-ID matcher (Tamburini, 2025, reporting the Luo 2019 metric) with preregistration, multiplicity deflation and mechanical held-out verdicts, released openly (§9.7), with Linear A worked as the honest null rather than a claimed crack.

3 Methods: the discipline harness

The harness is a comparison-layer pipeline that a reader can reimplement. A candidate reading enters as a *proposal* and passes through four ordered stages — pre-registration, deflation, decontamination, held-out graduation — under a cross-cutting persistence invariant: every decision is computed from stored artifacts, never from a model's narration. Model-derived signals — LLM theses, learned embeddings, cognate-matcher assignments — enter capped at confidence $\leq$ 0.75, structurally unable to *decide* a verdict; only mechanically-verified held-out reads rank higher (§1–2).

3.1 Pre-registration

Before any model is built or run, the predictions are frozen. The morphology pre-registration fixes, for each hypothesis, the claimed affix, its source, the falsifiable test, and the acceptance threshold, frozen in-repository before each run so the model cannot be tuned to them after the fact (the anchoring mechanics and the per-affix acceptance criteria are given in Appendix A.1). Outcomes are enumerated in advance — CONFIRM, EXTEND, REFUTE/NULL, or NO POWER — so that a negative is a publishable result, not a failed experiment. Any change is a new dated pre-registration, never an edit.

3.2 Deflation

Internal fit on the derivation set is treated as no evidence. Every match rate is deflated for everything tried: `n_hypotheses` × `n_signs` × `n_roots`. The nulls are pipeline-level — a frequency-banded permutation null (after Packard, 1974) and the order-statistic bar of §3.4 (implementation detail in Appendix D). Multiplicity is instrumented, not hand-passed, by a deduplicating `SearchLog` (invariant 12; Appendix D). The strong structural statistic, `S_morph`, earns credit only through a productivity gate — an affix must attach to $\geq$2 distinct residual stems, so a single formula word copied across inscriptions scores zero (calibration in Appendix A.2, Table 11).

3.3 Decontamination

A large open-weight model may have encountered longstanding published readings by Gordon and Best; reproduction of indexed historical proposals is therefore treated as potential shared-source contamination rather than independent evidence. The June-2026 Di Mino claim is analysed separately as a contemporary public case and is not assumed to be present in the model's pretraining data — it is indexed so that future claims and models cannot reproduce it as discovery. Three mechanisms address this.

The literature index records published Linear A sign-value and correspondence claims; any proposal matching the index is tagged `literature_match` and quarantined — it may be tested but can never count as discovery. The index partitions the sign inventory into `L_indexed` (referenced by $\geq$1 published proposal) and `L_not_indexed` (untouched). Discovery claims may rest only on `L_not_indexed` held-out success, since regurgitation can only return what is in the literature. (The conservative seed, the adversarial population, and the census are given in Appendix C.5, Table 12.)

The `L_fake` canary is a phonotactically-plausible invented lexicon, calibrated to a real candidate language (Appendix C.5), generated after the model's release and never published by us; exact-string collisions with real lexica are measured (Appendix C.5). Running the full pipeline against it yields a **generator-conditional** false-positive floor: any apparent lexical support it produces is spurious with respect to the target historical-language hypothesis, and memorisation cannot be invoked because there is nothing to memorise. A real candidate's match must outperform this *one calibrated class of negative controls* by the corrected margin — support means the candidate beat this generator-conditional control family (here a first-order sign-bigram / root-template generator), not that it defeated every possible false-language process.

LLM-ablation compares the pipeline with and without the model's proposals to isolate contamination (procedure in Appendix C.1).

3.4 The held-out graduation gate

A reading graduates only mechanically, from held-out data, as a conjunction of pre-registered clauses computed by `scripts/verdict.py` — never by the model that proposed the reading. The operative deflation clause is an **order-statistic bar**: the observed held-out statistic must exceed the *expected maximum* of the multiplicity-null taken over the number of trials the search was free to run (the E[max] over `n_trials` deflation), a **normal-order-statistic approximation** in which n = **`n_trials` is the *count of candidates the search tried*, not an estimated number of independent hypotheses**. The explicit formula and its Gaussian/independence assumptions are given in Appendix D; the error-control claim is carried not by the formula but by the direct empirical calibration below. This is a direct correction for selection across the hypothesis space, and it fails closed on a degenerate null (Appendix D). The framework yields four primary evidential outcomes: **GRADUATE** (every clause holds and the search multiplicity was instrumented), **REJECT** (a clause failed, or the statistic sits below the corrected bar), **NULL_PUBLISHED** (the statistic is indistinguishable from the `L_fake` null — a calibrated non-result, publishable as such), and **NO POWER** — the last assigned *upstream*, by the pre-registered module outcome when the experiment cannot adjudicate (§3.1), before the gate sequence runs; the gate itself additionally emits a fail-closed **INCOMPLETE** status when search multiplicity is un-instrumented. These outcomes concern *evidential support, not truth*: because Linear A has no ground truth, REJECT / NULL / NO POWER mean a reading is unsupported or insufficiently validated on held-out data, **never that the underlying linguistic proposition is false**. The verdict follows a fixed decision sequence (`verdict.py`): all pre-registered clauses hold on an above-null match → **GRADUATE**; else a match clearing the substantive clauses but on an un-instrumented search → **INCOMPLETE**; else an observed statistic not exceeding the null mean → **NULL_PUBLISHED**; else → **REJECT**. We calibrate the refusal rule directly: under the tested best-of-100 random-map null — an analyst trying 100 random candidate maps and submitting the best — 3 of 500 replicates graduated, an observed false-graduation rate of 0.6% (one-sided exact Clopper–Pearson 95% upper bound $\approx$ 1.54%), with 487/500 rejected outright and 10 published as calibrated nulls (Appendix D); this is an empirical error-control result conditional on that null-generating process, not a universal guarantee. A second pre-registered clause, **`generalizes_to_not_indexed`**, requires the reading to recover literature-unseen (`L_not_indexed`) signs — so reproducing already-published values cannot by itself graduate a claim; the clause fails closed (its pre-committed thresholds are in Appendix D). Graduation is defined as validated *novel* discovery; independent held-out support for an already-published reading is recorded as a distinct outcome and does not graduate. The remaining pre-registered clauses (`L_fake` separation, corrected margin, order-statistic bar) are conjoined as specified in Appendix D. These are *gate parameters* — pre-registered choices for this corpus, not universal constants. Validation is leave-one-site-out, not ordinary *k*-fold (Appendix A.1). The gate is AND-monotone — clauses can only make it stricter — so hardening never lets a previously-failing reading graduate.

3.5 Grade from persisted artifacts

The invariant across every gate: verdicts are computed from persisted rows, never from a model's account of its own outcome. The verdict writer is grep-clean of any model call and is the sole writer of the verdicts table.

The same falsification-first harness returns *different* calibrated verdicts across probes (Table 1) — the paper's contribution in one view. Each number is detailed in the cited section.

4 The information floor

Before any decipherment method is graded, the corpus must be sized against what it can identify. Two quantities matter: how much text exists, and how many free parameters a reading spends against it. Confusing the first for the second is the origin of the "too little text" folk claim — and, we argue, of its equally unearned inverse, the optimism that a big-enough corpus makes a reading identifiable.

The corpus. The current authoritative synthesis (Salgarella, 2025) gives an envelope of roughly 2,500 total Linear A finds, of which about 1,400

Probe	Control / comparator	Real outcome	Null / control floor	Verdict
Graduation gate (§3.4)	—	—	best-of-100 random-map null	3/500 false graduations (0.6%; calibrated under the tested null)
Morphology (§6)	Linear B 9/16	Linear A 4/16	`L_fake` bigram 6/16	NO POWER
Segmentation (§6)	—	micro-F1 0.436	random 0.389	supported (gap CI excludes 0)
Metrology (§7.1)	planted 0.9	0.000	permutation 0.113	null (agrees with the field)
Phonology (§7.2)	planted (fires ≥ 2)	0.080 / 0.140	0.065 / 0.201	null (data-limited)
Image alignment (§7.3)	font-swap survives	0.410	image chance 0.0135	circular demonstration (capped)
Contamination (§8.1)	model-free 0/40	14/40 (JA-NE)	0/40	exploratory (public-exposure gradient)

Table 1: One harness, calibrated epistemic outcomes across probes.

are readable inscriptions, overwhelmingly concentrated in the single LM IB destruction horizon (findplace and dating detail in Appendix C.4). That near-single horizon is why a site-held-out split, not a chronological one, is the only realistic partition. No bilingual text has ever been found (Salgarella, 2025, §8).

Corpus sizing and the unicity computation (numeric detail in Appendix C.4). The recurring sign-count confusion resolves into several distinct sign-count denominators and two occurrence channels that must be kept distinct (Table 10; Appendix C.4). On the parsed syllable-sign budget, Shannon's unicity distance sits everywhere far below the corpus size (Table 10; Salgarella, 2025, §1–3); we retain this only as a toy-model *precondition* diagnostic and explicitly withdraw it as any sufficiency or refutation claim. Qualitatively, the figure cannot settle identifiability either way (Appendix C.4): unicity distance presupposes a known plaintext-language oracle we do not have. Corpus size, in short, cannot decide the question.

What actually binds. The results are consistent with *identifiability* being the more fundamental present constraint rather than corpus size (Salgarella, 2025, §8); we do not establish that more data would be irrelevant. It has three parts: (i) the **absence of a known cognate language** — the map has a value-space but no anchored target (Luo 2019's limit); (ii) the **multiple-testing / search trap** — that a unique consistent map may exist in principle neither guarantees an algorithm finds it nor stops a cherry-picked false positive across the parameters tried (invariant 8); and (iii) a **mixed inventory** of syllabograms, logograms, and numerals that a real reading must first separate. These, not the count of surviving signs, are what a decipherment must overcome. Accordingly, when a candidate lexicon grows without bound on a fixed occurrence budget — free parameters exceeding the floor — the reading is underdetermined, and the framework must say so.

5 A qualitative audit of the *301 claim

The framework of Section 3 is abstract until it meets a concrete claim. In June 2026 a widely circulated "Linear A cracked" announcement (Di Mino, 2026) argued that Linear A records an extinct Semitic language, reviving Gordon's hypothesis (Gordon, 1957, 1966). Its load-bearing anchor is a single reading: the recurring peak-sanctuary libation formula word A-TA-I-*301-WA-JA, in which the Linear-A-only sign *301 is *itself* assigned a /na/-initial value, so the verb root reads as West-Semitic N-W-Y "to dwell" (*nawaya*) and is thereafter matched to later Hebrew liturgy. We analyse this **as a claim, on its public evidence** — not the person — from an immutable archived copy carried in the artifact. It is a clean, self-contained instance of the failure modes the gate is built to catch, and the three-part refutation below rests wherever possible on the claim's **own published table** and on primary sources (Davis, 2013; Duhoux, 1992; Thomas, 2020; Salgarella, 2025; Steele and Meissner, 2017) — none advanced in logos's own voice, and the paper introduces no new sign reading. We flag the meta-point once: this is a literature-and-primary-source argument, not itself a logos held-out result — the graduation gate and `L_fake` canary are validated as instruments (unit tests; the known-answer Ugaritic↔Hebrew surrogate) but have **not been run on this claim**. Scoring Gordon's equations and the Di Mino lexicon against the `L_fake` floor and the order-statistic bar is directly runnable with the components of §3 and is the priority of the planned revision; that the refutation has not been independently vetted by Aegean epigraphers is a disclosed limitation (§9.6), not a standard the work claims to meet.

(a) The segmentation is not novel. Root i-*301 plus affixation is the reading of Davis (2013); an independent segmentation by Thomas (2020) reaches the same core and is tested head-to-head against it (the current synthesis, Salgarella, 2025, does not itself adopt Thomas). On the Davis and Thomas segmentations, the affix-stacking is evidenced by multi-attestation forms — ta-na-i-*301-u-ti-nu (IO Za 6) and a-ta-i-*301-de-ka (ZA Zb 3) — that vary the material around a stable i-*301 core. (These sigla are confirmed against Davis 2013: `TL Za 1`, `IO Za 6`, and `ZA Zb 3`.) The claim re-segments a structure already established in the literature. Under the decontamination layer this matters procedurally, not just historically: a correspondence that reproduces a published segmentation is treated as potential shared-source contamination, tagged `literature_match` and barred from counting as discovery (the artifact, §C.1).

(b) The gloss is contested and weaker. The mainstream value of the verb is "give/dedicate" — Duhoux's (1992) gloss, adopted by Davis (2013) as a *structural* reading that assigns *301 **no phonetic value at all**, and endorsed by the current synthesis (Salgarella, 2025, §7.2). "Dwell" is a semantically distant, unmotivated substitution. The point is not that "give" is certainly right, but that a *structural* reading needing no sound-value for *301 already accounts for the data the phonetic claim is invoked to explain.

(c) The primary descriptions of this form's morphology do not support the Semitic label. The claim's own table segments the verb as prefix-stacking and agglutinative — [prefix A "I"] [stem-morpheme TA] [stem-vowel I] [root *301]. We deliberately do **not** rest on an *a-priori* typological contradiction here, and we retract the earlier draft's claim that a prefix-stacked verb is "internally incoherent" with a Semitic reading: Semitic verbal morphology includes productive prefix conjugations, so a prefix on the verb is not by itself un-Semitic. The point is empirical, not typological, and it comes from the primary literature. Davis (2013, p. 38 n. 13) characterises i-*301's morphology as "neither IE nor Afroasiatic" — and Semitic *is* Afroasiatic, so a decade earlier the primary description already found **no Afroasiatic morphological signature** in this very form. (We rest on what the footnote states — the absence of an Afroasiatic signature — not on a stronger "ruled out" than a hedged footnote can bear.) Independently, Salgarella (2025, §8) concludes Minoan is agglutinative (after Duhoux 1978) and a language isolate. The claim's own segmentation is thus consistent with the isolate typology, while the specific Semitic assignment finds no morphological support in the primary descriptions of the form.

The free-parameter receipt. The decisive point for identifiability is that *301 has no cross-script anchor: in the claim's own table its AB-number column is "—" and its Linear B correspondence "(none)". Even the standard practice of reading Linear A homomorphs with their Linear B values is a backward projection to be handled with care (Steele and Meissner, 2017, already cautioned by Duhoux 1989, apud Davis 2013, p. 38 n. 11) — and an A-only sign such as *301 has no homomorph to project from at all, so nothing cross-script constrains its value. The single sign on which the entire decipherment pivots is thus a **free parameter** — its value was assigned to fit the desired root, not constrained by any independent evidence. This is precisely the case the minimum-description-length / identifiability *diagnostic* flags: a value chosen to make one word read out carries no description-length saving that survives the multiplicity of alternatives it was selected from (cf. Section 3; the diagnostic is reported next to every claim, and the operative graduation clause — the order-statistic bar over exactly that multiplicity — is one a hand-fitted value cannot clear). For this reading the obstacle is identifiability rather than corpus size — no known cognate language to constrain the assignment, uncorrected multiplicity, and a sign whose value is chosen rather than anchored (a free parameter is unanchored at any corpus size).

Read against Section 3, the claim instantiates two distinct failure modes at once. It is **conclusion-driven model specification** — the table's *structure* says agglutinative and prefixing, but the *label* says Semitic, and the label wins because the desired conclusion (a Semitic prayer) is a fixed prior rather than an emergent reading. And it is a **free-parameter / identifiability** failure — an unanchored sign-value tuned to the target, on a corpus with no known cognate language to constrain it. logos indexes the reading (*301 = /na/) precisely so a model cannot regurgitate it as discovery, and quarantines rather than dismisses it. The lesson generalises: a match rate on a hand-picked word, absent a committed prediction and multiplicity accounting, is not evidence. The *301 claim is a qualitative illustration of the failure modes the gate is designed to detect.

6 Results I: morphology induction

We test whether Linear A's pre-registered affixes and word divisions reflect genuine morphology. The apparent affixation does not clear the operative `L_fake` bigram floor, so the verdict is NO POWER and `has_validated_morphology` = False. The identical test **validates on Mycenaean Greek (Linear B)**, whose longer words carry uncontested inflection: it confirms affixation at rate 0.562 — above the same bigram floor ($\sim$0.30) — with `has_morphology_power` = True, so the control rules out a universally insensitive detector and supports the reading that Linear A's shorter forms contribute materially to its lack of power (Table 4; Appendix A.1). One held-out positive stands and survives its own uncertainty: a DP-unigram segmenter recovers the scribe's own word divisions at micro-F1 0.436 versus a random-boundary baseline of 0.389, a gap whose site-clustered 95% bootstrap CI [0.021, 0.099] excludes zero (P(gap $>$ 0) = 0.998; Appendix A.1). This separates real recoverable structure from fitted morphology — the paper's core discipline.

7 Results II: metrology, phonology, and the cross-script asymmetry

Three further pre-registered probes exercise the harness against distinct field claims. Each returns a calibrated null or a truth-capped positive; together they map where the Linear A corpus does and does not carry recoverable signal, and they are consistent with identifiability rather than effort being the more fundamental present constraint.

7.1 Metrology: a parse-coverage null that agrees with the field

Direction D poses the accounting tablets as a constraint-optimisation over the fraction-sign values (Appendix B.1; Table 5). Held-out fraction balance does not beat the shuffle null (p = 1.0), so the verdict is NULL; the only value the method determines, the fraction sign J = 1/2, is the long-standing field consensus (Bennett, 1950), not a logos discovery. This is a null the field itself reached (Corazza et al., 2021) — the harness crediting a surviving value to its originator rather than rebranding it.

7.2 Distributional phonology: a data-limited null bounded by a power curve

Does a sign's distributional context (which signs sit beside it) carry information about its phonetic value? Tested on the known AB signs, neither the consonant class nor the vowel class clears chance (permutation p = 0.40 / 0.86). The verdict is a data-limited NULL: the sound path is not recoverable from Linear A alone, and no sound value is imputed for any sign (Table 6; Appendix B.2).

7.3 The cross-script asymmetry: image alignment recovers shared anchors, but the recovery is circular by construction

Do image embeddings recover the cross-script A$\leftrightarrow$B correspondence that sequence and cognate co-occurrence cannot? Classical shape descriptors reach direct-NN **0.410** on held-out A$\leftrightarrow$B anchors while sequence and cognate co-occurrence stay at chance. But the recovery is circular by construction: the A$\leftrightarrow$B anchor values were themselves assigned by twentieth-century epigraphers on sign-shape similarity, so this is a shape-genealogy engineering demonstration, truth-layer-capped $\leq$ **0.75**, not a positive decipherment claim (§9.6; Table 7).

8 Results III: contamination probes

One failure mode is invisible to the false-positive / multiplicity axis of §7: a decipherment that *looks* novel but silently reproduces published readings a language model has memorised. This section reports the two decontamination probes that target it — both run as method demonstrations, returning completed (if deliberately bounded) findings. Throughout, where a clean number was not obtainable we report the partial-null and the confound rather than a headline.

8.1 LLM-ablation: does a large open-weight model regurgitate the literature?

In the LLM-ablation, `qwen2.5:72b` reproduces famous published readings — Gordon's JA-NE = "wine" in 14/40 seeds — that the model-free edit-distance baseline never reaches (0/40), yet never reproduces obscure vessel equations (0/40; Table 8). Verdict: a reproduction gradient consistent with differential public exposure — though the experiment does not isolate memorisation from correlated item-level differences — the memorisation confound the discipline layer must gate out.

8.2 `L_not_indexed`: indexed versus not-indexed abstention

Does the model's behaviour on *published* signs generalise to signs unseen in the indexed literature? `qwen2.5:72b` valued the memorisable known signs in $\sim$20/40 seeds but abstained $\sim$97% of the time on the literature-unseen *-series signs ($\sim$1/40; Table 9) — an abstention asymmetry consistent with differential public exposure (the §8.1 caveat applies). With no ground truth on the not-indexed signs, **consistency is not correctness**: it bounds how the model behaves, never whether it is right.

Registered extension: a known-answer sufficiency protocol (asserts no result)

Is a Linear-A-scale corpus large enough to recover a decipherment if a known cognate existed? Because every benchmark is handed a real cognate signal Linear A lacks, any recovery curve is an **optimistic benchmark, conditional on a known cognate relationship**, readable two ways: at-floor at Linear-A scale ⇒ information-insufficient even given a cognate; above-chance ⇒ identifiability (cognate-absence, uncorrected multiplicity) rather than corpus size would be implicated. Neither branch is ever a decipherability claim. The accuracy-vs-size curve is deposited in the repository as a known-answer upper-bound **protocol** (Appendix C.3); this paper asserts no interim result from it.

9 Discussion, limitations, and conclusion

9.1 The discipline is the contribution

The deliverable of this work is not a reading of Linear A. It is a *harness* — a literature index with an indexed versus literature-unseen partition, a fabricated-language control corpus, model-ablation contamination checks, multiple-comparison correction, and pre-registration, all graded mechanically from persisted artifacts — together with a body of pre-registered, calibrated nulls, each with its borderline case named. The stratified morphology re-run is the template: at the pre-registered $\alpha = 0.01$, no pre-registered affix is cross-stratum stable above the bigram floor, robust to the genre-bucketing choice (Tables 2–3; Appendix A.1). A null that names its threshold, its sensitivity, and its near-miss is a stronger referee object than a flat negative.

The axis this harness targets is the **false-positive / multiplicity axis** — the mechanism by which a determined analyst "translates" almost anything on a small corpus — a property of inference methodology, not of the object or its data (§1–2).

9.2 Two named failure modes for the methods literature

(i) Narrative capture. Structure says X; the label says the desired Y; the label wins. In the June-2026 *301 claim the published table segments an agglutinative, prefix-stacking verb while the assigned family label is Semitic, and the desired output operates as a fixed prior, so the conclusion conforms to the prior rather than emerging from constrained readings (§5). This is distinct from statistical overfitting and must be named separately. Notably, the mismatch is empirical — between the label and the primary descriptions of the form (§5) — not an a-priori typological impossibility.

(ii) Free parameters exceeding the information floor. A lexicon reported as "408 terms and counting" on a fixed corpus has more degrees of freedom than the data can identify; a decipherment whose lexicon grows weekly is underdetermined by construction. The minimum-description-length / unicity *diagnostic* (`k ≤ U_floor`) is built to flag exactly this (it is reported next to every claim rather than enforced as a graduation clause; §4): the single sign the whole reading pivots on is the textbook free-parameter case (§5), its value assigned to fit the desired root; the refutation rests on the claim's own published table, archived in the artifact.

9.3 The post-hoc evaluation sequence

These failure modes trace to sequencing. The *301 claim publicly commits to a post-hoc evaluation sequence: declare "officially deciphered," then vet later. logos's documented inverse is **pre-register → held-out → locked artifact** (§3.1, §3.4; the held-out libation-formula corpus and its low-power caveat are in Appendix A.3).

9.4 Identifiability appears the more fundamental present constraint

We take care not to write a defeatist framing, and equally not to over-claim from corpus size: the toy-model unicity distance ($U \approx$ 204–415 signs; Table 10, §4) is a *precondition* diagnostic only, explicitly withdrawn as any sufficiency or refutation claim — "$U \ll N$" does **not** license the statement that there is enough text to pin a decipherment (§4; Appendix C.4). The results are consistent with *identifiability* being the more fundamental present constraint — the absence of a known cognate language to map to, an uncorrected multiple-testing search burden, and a mixed sign inventory — rather than a demonstrated corpus-size threshold; a null can arise from identifiability, low power, or size alike, and we do not establish that more data would be irrelevant. Salgarella (2025, §8) reaches the field-level version of this by a different route (§2.2).

9.5 Study-wide error budget

Multiplicity discipline operates at two levels, and we are explicit about both. Within a single probe, matches are deflated for everything searched (§3.2–3.4). Across the study, the unit of multiplicity is the **per-probe pre-registration**: the **five executed probes** (morphology, metrology, phonology, cross-script image alignment, and contamination) each carry their own pre-committed α, and every probe is reported regardless of outcome. **Error control is defined within each pre-registered probe; we claim no global family-wise guarantee across probes.** Nothing in the battery is read as a decipherment.

9.6 Limitations

The nulls are honestly bounded, not decisive. A dependence-adjusted trial-count correction of the morphology z-test is deferred; the deferral would only strengthen the null (Appendix A.1). No held-out *decipherment* has been achieved, and the study reports **no positive decipherment claim** — the deliverable is calibrated absence, not a reading. Nothing here claims a rejected reading is *false* (§3.4): every REJECT / NULL in this study bounds *evidential support* on held-out data, not truth. The one probe that recovers an above-chance number, cross-script sign-image alignment, is a font-robust but circular *engineering demonstration* (a font-swap control refutes the typeface-artifact worry; §7.3, Appendix B.3), truth-capped at $\leq$ 0.75, never a reading. No phonetic claim is made. The philological points in §5 rest on the published literature cited there rather than on independent Aegean-epigraphic review, which we disclose as a limitation; because those points are cited rather than original, the argument does not depend on any new sign reading of our own. Chronological cross-validation is largely foreclosed by the single-horizon corpus (§4; Appendix A.1). And the corpus itself — small, partly lossy, with no bilingual yet found (Salgarella, 2025, §8) — sets the ceiling.

9.7 Conclusion and reproducibility

The framework withholds validation when a proposed reading does not clear the pre-registered gates (§1, §3); the contribution is the framework and the calibrated null itself — the instrument and the honest limits of what the corpus can support — not a reading, and the calibrated null is a safeguard against unsupported claims. The code, pre-registrations, and all findings are open under the MIT licence and available as an anonymized artifact (code and pre-registrations released on acceptance). The licensed epigraphic corpora are not redistributed; each source's licence and retrieval path is documented (data-provenance; Software & data note), so the harness is reproducible against a locally obtained corpus. This work is independent and not affiliated with, or endorsed by, any of the cited projects or their authors.

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A Morphology (replication detail)

A.1 Morphology: rates, floors, word-length, stratification, sensitivity. The probe is pre-registered, unsupervised morphology and segmentation induction over the full Linear A corpus (Table 10). Target claims are the recurring affixes and positional grammar of the Davis/Thomas/Duhoux–Valério tradition: prefixes and suffixes (`i-`, `a-`, `-te/-ti`, and Davis's `i-⋆301` verb stem). Pre-specification anchoring (§3.1): pre-specifications were frozen in-repository before each run (commit history); the external immutable deposit ([artifact DOI withheld for review]) certifies the current state of the registrations, not temporal priority. Each pre-registered affix (drawn from Thomas (2020) and Duhoux/Valério) must recur on $\geq$2 distinct stems across independent inscriptions, above a within-form permutation null, and survive multiplicity correction at a target of $p < 0.01$ corrected. Every candidate was graded against two null floors: a within-word sign-order permutation (shuffle) floor, and an `L_fake` floor built from a first-order sign-bigram (Markov) corpus with zero lexical overlap with the real inscriptions.

Pooled run. Of the 16 pre-registered affixes, 4 beat the within-form permutation null and the deflated multiplicity bar (`i-⋆301`, `a-`, `i-`, `-te/-ti`), clearing the shuffle floor decisively (rates and Wilson intervals in Table 4). The apparent standout `i-⋆301` sits far above the permutation null — but that figure is computed against the weaker within-form permutation null only, and it does not survive the operative `L_fake` bigram floor, under which the whole set fails: the fabricated Markov corpus confirms the same affixes at a *higher* rate than the real corpus (Table 4). The outcome is classified NO POWER, not evidence against morphology; `has_morphology_power` is False and the module reports the null (the Linear-B positive control that validates the detector is reported below).

Word-length distribution. Linear A words are short (Table 4), and on such words "morphology" and "bigram order" are statistically indistinguishable. This reconciles with Fuls's 3.3-sign figure once the denominator is matched: Fuls (2015, 57, §6.5) computes the mean over a word *list* of "554 complete sign sequences, where duplicates are reduced to one" (GORILA; Godart and Olivier, 1976–1985) — distinct word *types*, not tokens; restricting our corpus to distinct multi-sign types likewise recovers 3.07. Fuls reads the 3.3-sign length as evidence that Minoan is *polysynthetic*; our direct test finds the apparent affixation bigram-reproducible with no power, and we report the null. The direct positive control — the identical bigram-floor test on Mycenaean Greek (Linear B), whose longer words carry uncontested inflectional morphology — was RUN on the DĀMOS corpus (parsed by the same `_damos_wordforms` extractor as §7.3) against a pre-registered 16-affix Mycenaean panel (Ventris and Chadwick, 1953). It fires (Table 4), carried by the securely-grammatical suffixes -jo, -ojo (gen sg -oio), -de (allative), -qe (enclitic "and"), -o-i (dat pl), so `has_morphology_power = True`. The identical test that returns NO POWER on Linear A's short words recovers Mycenaean morphology on longer ones (a sensitivity check: an ad-hoc parser yielding mean 2.19 signs/word did *not* fire — power tracks word length exactly as the short-word argument predicts), which rules out a universally insensitive detector and supports the interpretation that Linear A's shorter forms contribute materially to its lack of power (`scripts/comparison/linb_morphology_control.py`).

Segmentation positive. Under leave-one-site-out cross-validation (52 sites; not k-fold, given formulaic dependence), a DP-unigram (Goldwater-style) segmenter recovers the scribe's own word divisions above a random-boundary baseline; a site-clustered bootstrap (resample the 52 sites with replacement, 4,000 draws) puts the real-minus-random gap above zero, and the paired gap, not the wide marginal per-segmenter CIs, is the operative test (Table 4). Morfessor over-segments the short syllabic corpus and is reported as not usable rather than hidden (Table 4).

Stratified re-run. The strata reconcile exactly to the 1,341-inscription corpus total (Table 2). The 99-inscription "libation" *genre* stratum is not the same object as the libation-*formula* corpus (§A.3). Following Salgarella and Judson (2024), chronological cross-validation was explicitly declined (988/1,341 $\approx$ 74% single-horizon LM IB), and site — not scribal hand, which is not individuated for Linear A (Salgarella, 2019) — is the independence unit, gated by presence at $\geq$2 distinct sites. The result is `has_validated_morphology` = False (§9.1). Stratification also showed that even `i-*301` is bigram-reproducible in the repetitive libation register, where the Markov corpus manufactures the apparent verb morphology; Davis's slot grammar is not refuted, merely not separable from sign bigrams here. Where formula-bearing sites drive a signal, power is intrinsically low: a leave-one-site-out CV over so few formula findspots (an effective $\sim$5-fold site partition) cannot resolve a modest effect.

Sensitivity, disclosed. The honest 2$\times$2 over genre bucketing $\times$ α threshold is Table 3; the doubly-relaxed corner's `i-` (locative) and `-te/-ti` (ablative) are precisely the Duhoux/Valério H3 nominal affixes that Salgarella (2025) §8 independently endorses — the named borderline candidates, fragile to *both* the α threshold and a defensible genre choice (the $\alpha = 0.01$ null is itself robust to the bucketing). The present analytical approximation uses attempted-candidate count rather than a dependence-adjusted effective trial count — a recorded, deferred limitation whose correction would only strengthen the null.

A.2 `S_morph` detector calibration (§3.2). After the productivity-gate fix (§3.2 — an affix must attach to $\geq$2 distinct residual stems), `S_morph`'s measured false-positive rate on within-form-shuffled corpora is consistent with the nominal one-sided $z \geq 2$ rate within sampling error, and its planted-affix power is a saturated point, not a full sensitivity curve (Table 11; contrast the phonology power curve, §7). This is an **indicative calibration check, not the operative gate**: 6/200 is too coarse to certify an $\alpha = 0.01$ threshold on its own, and the gate applies the order-statistic bar over the far larger permutation and candidate counts the deflation machinery tracks (§3.4).

A.3 Held-out design: the libation-formula corpus (§9.3). A leave-one-site-out split over so few formula sites (§1) has low statistical power — and site-level CV is in any case necessary, not sufficient, since within-site scribal-hand correlation must also be controlled (Salgarella and Judson, 2024). This formula corpus should not be conflated with the 99 stone-vessel inscriptions of the "libation" genre stratum (§6): they are different objects.

B Metrology, phonology, and cross-script (methods and controls)

B.1 Metrology. Direction D parses HT (LM I) documents into balance units `[recipient][commodity logogram][integer][fraction signs]` and solves the fraction-sign values by RANSAC over exact-`Fraction` Gaussian elimination (robust to the unbalanced-tablet outliers that defeat least-squares), requiring line items to balance the preserved KU-RO totals; validation is grouped k-fold by document (no tablet straddles train and test) against a permutation null. Results are in Table 5: of $\sim$7 fraction-bearing constraints only one is satisfied by the solved system, the sole determinable value is J = 1/2, held-out fraction balance is not separated from the permutation null, and the synthetic positive control (planted values, all recovered) separates. The null traces to sparse automated parse coverage (multi-commodity blocks, illegible number glyphs, KI-RO deficit sub-lists, editorial restorations); Corazza et al. (2021, p. 2) abandoned the balance-to-totals method for the same reason. The surviving J = 1/2 (§7.1) gains Schrijver's (2014) three further tablets (HT Zd 156, PE 1, HT 123a): at least four corroborating documents.

B.2 Distributional phonology. Track 2 asks whether a sign's distributional context — which signs sit beside it — carries information about its phonetic value, measured on the known `AB` signs whose Linear B sound values can be borrowed. Features are PPMI-weighted, L2-normalised left/right co-occurrence counts; labels are the Linear-B-transferred (C, V) parsed from the GORILA name, so features never see the labels. The test is leave-one-out 1-NN against a label-permutation null, Šidák-deflated over the two tests; 50 of ∼55 AB signs met the ≥3-occurrence threshold.

Per-test results are in Table 6. The honest verdict is *data-limited*, not *no bridge exists*, because the positive control disciplines the reading: the positive-control power curve over planted strengths shows the test firing at strength ≥ 2 but not at 1. The test works but is not infinitely sensitive at n = 50 / 13 classes, so we cannot distinguish "no distribution→phonetics bridge" from "too few signs to detect a weak one" (§7.2, §B.3).

B.3 Cross-script image alignment. The palaeographic probe renders 88 Linear-A and 74 Linear-B glyphs and tests held-out recovery of shared A↔B anchors. Classical descriptors (HOG + Hu moments + shape-context) sit far above image chance, and a from-scratch I-JEPA is excluded from the claims as over-parameterized for ∼90 glyphs (results and controls in Table 7). Two controls decide how to read the image-vs-sequence asymmetry (§7.3), and together they *downgrade* it from a positive to a demonstration.

First, a font-swap control. Rendering both scripts in one typeface (Aegean, George Douros) risks making the alignment a single-designer house-style artifact rather than a fact about the signs. We re-render Linear A in Noto Sans Linear A and Linear B in Noto Sans Linear B — two independent type designers, each font covering only its own Unicode block (Noto-A carries zero Linear B glyphs and vice-versa), so no single hand draws both scripts. The recovery survives and is if anything stronger (Table 7; both legs ≫ chance): the font-artifact hypothesis is *refuted*, and "distances reflect shape, not font" is now demonstrated rather than asserted (`scripts/palaeo/font_control.py`).

Second, and decisively, the circularity the font control cannot remove. Linear B was historically adapted from Linear A, and the A↔B transcription *values* that define the anchor set were assigned by twentieth-century epigraphers largely on the basis of sign-shape similarity, so the recovery re-derives a shape-defined correspondence (§7.3): it confirms a known palaeographic fact (borrowed signs look alike) and validates a font-robust descriptor pipeline, but it reads no Minoan. It remains truth-layer-capped in any case — the Ferrara et al. (2021) "Archanes Formula" hard negative (`CH 095` resembles `AB 60`/*ra* but is not its phonetic counterpart) shows a plausible shape match can be phonetically wrong, so a shape match is a ≤ 0.75 signal (invariant 5), never a reading. The honest asymmetry is thus: the image path re-derives a shape-defined mapping (expected, circular, capped), while the sequence and cognate paths **remain at chance even after ingesting the full DĀMOS Mycenaean corpus** — 13,562 wordforms, ≈15× the earlier cognate file, 56 anchors (Table 7) — while the *positive control recovers*, so the harness works and the null is a real null: the A↔B phonetic correspondence is simply not carried by cross-script sign co-occurrence, even at full Linear-B scale (§B.2). And the A-only signs on which positive-decipherment claims would pivot have no cross-script anchor at all (§5), so images cannot impute them even in principle.

C Contamination and sufficiency (detail)

C.1 LLM-ablation reproduction gradient. The ablation (§3.3) is the remaining gate on the contamination number, matching a model's cognate glosses against the indexed lexical claims. We ran it against `qwen2.5:72b` — a large open-weight model, a far stronger memoriser than a local 12B — on a rented H100, with an NW-Semitic candidate and a Hebrew skeleton lexicon for the model-free arm (scale and quantities in Table 12) (the artifact). The first-pass contamination rate is an **artifact, reported as such**: the mechanical arm never shares a literature-matching correspondence with the LLM (their value vocabularies barely overlap), so the rate is forced to 1.0 by construction and is not reported as a contamination result (Table 12; the artifact).

Generation settings (replication). Checkpoint `qwen2.5:72b` as served by Ollama (Ollama's default `q4_K_M` quantization; Ollama version not recorded), on an ephemeral vast.ai H100 SXM 80 GB instance (torn down after the run). One `/api/generate` call per seed, stream off, timeout 180 s; temperature 0.8; sampling seed = the replicate seed; top-p, top-k, maximum output length, and stop conditions left at server defaults (not recorded). Each seed's 40 held-out forms are drawn deterministically (`numpy.default_rng(seed)`) and rendered as canonical sign names in a fixed JSON-request prompt (`_PROMPT_TEMPLATE` in `scripts/comparison/ablation.py`). Parsing: the JSON `partial_map` is extracted; keys not naming a sign shown to the model are dropped (the proposer cannot invent signs); a dead or garbled call contributes an empty proposal without crashing the batch (fail-closed). The gate-2 rerun prepends the 7 indexed probe forms to every seed. Per-call logs: `runtime/ablation/llm_calls.jsonl`; harness commits `0582966` (ablation) and `9cd6d04` (gate-2).

The gate-2 rerun, reconciled to a common consonant-skeleton vocabulary, yields the honest deliverable: the per-item reproduction gradient over 40 seeds (Table 8; the artifact). An external red-team correction split `KU-RO` between the Semitic-root route (*kull*) and the general-knowledge meaning "total" (confirmed by tablet arithmetic), so `KU-RO` = "total" is **excluded** as a witness and `KU-RO` = *kull* is downweighted (Table 8; the artifact). The surviving gradient — famous readings reproduced, obscure ones not — is what the ablation detects (§8.1). The gate-2 rate itself remains confounded by the baseline's representational ceiling and is not reported as a contamination measure (Table 12).

C.2 `L_not_indexed` abstention asymmetry. The `L_not_indexed` probe shows a sign in real context (§8.2); the not-indexed signs are absent from the indexed sources, hence not memorisable from them (never "impossible to have seen" in any absolute sense) (the artifact). Across 40 seeds, `qwen2.5:72b` abstains ∼97% of the time on the untouched signs while confidently valuing the memorisable ones (Table 9). The apparent known−not-indexed consistency gap is coverage-confounded and was flagged, not asserted: the matched-*n* robustness check returned `no_power`, so the rigorous comparison could not run (Table 9). The interpretable signal is the abstention asymmetry itself, read under the standing rule of §8.2 (the artifact).

C.3 Information-sufficiency CSA protocol (design; asserts no result). The protocol down-samples known-answer benchmarks — Linear B / Mycenaean Greek as the primary syllabic analog, sourced through DĀMOS, and further syllabic corpora — and measures held-out cognate-ID recovery against per-size permutation floors, using the published cognate-ID matcher of §2.3 (Tamburini, 2025), validated in the artifact against its published Table-3 values (the artifact). The reading rules and the no-result stance are in the body (Registered extension). The phonological / sound path is no longer data-deferred: even the full ingested DĀMOS corpus leaves the cross-script distributional alignment at chance (§7.3, §B.3).

C.4 Information floor: corpus denominators and the unicity computation (§4). The corpus envelope, the four sign-count denominators, the two occurrence channels, and the measured unicity range are tabulated in Table 10 (Schoep 2002; Salgarella, 2025, §1–3, §6); the envelope spans ca. 1800–1450 BCE (Middle Minoan II–Late Minoan IB; §4). Each denominator must be labelled where it is used (Table 10). The channel through which structure is observed is a *sign-occurrence* count, not a document count or a sign inventory (invariant 7): the broader ∼7,400 occurrences count every sign type, while the narrower $N \approx 5{,}792$ parsed syllable-signs — after logograms and numerals are stripped — is what the unicity computation consumes. Two further properties tighten the budget: Linear A words are short and formulaic (the six-position libation-formula template; Salgarella, 2025, §7), and a single language/horizon dominates.

On the logos-normalized corpus, Shannon's unicity-distance criterion yields $U \approx$ 204–415 signs across the whole plausible (V, P) parameter space, everywhere far below N (Table 10) — retained only as a toy-model precondition diagnostic (§4). Unicity distance presupposes a *known plaintext-language oracle*; because we estimate the redundancy D from the ciphertext itself, that oracle collapses into Linear A up to relabelling, so the computation demonstrates only measurable recurrent structure, not that any reading is identifiable; and a lossy, non-injective sign→sound channel may possess no unicity distance at all (§9.4).

C.5 Literature index and `L_fake` canary (§3.3). *Index seed and adversarial popula-*

tion. The seed is deliberately conservative — a sign wrongly omitted is wrongly granted `L_not_indexed` status, the dangerous direction — and every verifiably-published disputed entry is indexed to *catch* regurgitation, never as an accepted reading. Population is adversarial: of the seeded West-Semitic lexical proposals, five are Gordon's equations and a-sa-sa-ra-me = "oh Asherah!" is Best (1981), not Gordon; provenance-unverifiable candidates — `qa-pa` = kappu and a Semitic `ki-ro`, whose attribution to Gordon is not supported by the primary account (Rendsburg, 1996) — were excluded rather than indexed. Exposure indexing is independent of evidential validity — a false public claim can still contaminate. Generated from the seed (invariant 12), the current census (Table 12) is a deliberately conservative, non-exhaustive seed, not yet the full §C.1 literature census. *`L_fake` calibration.* The canary is calibrated to a real candidate language's phoneme frequencies, root-template structure, and lexicon size (§3.3). Its own divergence from the calibration targets is reported, never minimised; with a Semitic root-template sampler and an external Hebrew (ETCBC/bhsa) reject set, the root-template total-variation distance fell markedly (Table 12), and any residual real-lexicon collision is measured and published rather than asserted to be zero.

D The order-statistic graduation bar: formula, assumptions, calibration

The operative deflation clause (§3.4) is a normal-order-statistic **approximation** to the expected maximum of the multiplicity-null over the attempted candidates. For a null of mean μ_0 and standard deviation σ_0 and n attempted candidates,

$$E[\max] \approx \mu_0 + \sigma_0\big[(1-\gamma)\,\Phi^{-1}(1-\tfrac{1}{n}) + \gamma\,\Phi^{-1}(1-\tfrac{1}{ne})\big]$$

with $\gamma = 0.5772\ldots$ the Euler–Mascheroni constant — the blended expected-maximum expression used by Bailey and López de Prado (2014); the general order-statistics treatment is David and Nagaraja (2003, *Order Statistics*, 3rd ed.). The pipeline exposes the bar as `expected_max_order_stat`(μ_0, σ_0, N); the deduplicating `SearchLog` (§3.2, invariant 12) reports the exact distinct-candidate count `n_trials` a scanner produces. The bar fails closed on a degenerate null ($\sigma_0 \le 0 \Rightarrow E[\max] = \mu_0 \Rightarrow$ no graduation). The `generalizes_to_not_indexed` clause (§3.4) requires recovery above a pre-committed threshold and on a minimum number of distinct not-indexed signs, and fails closed when no threshold or no supported not-indexed sign is present. The full clause conjunction: the candidate must separate from the `L_fake` negative-control distribution, its corrected margin must be cleared, and the observed statistic must beat the order-statistic bar over the Packard/random-lexeme multiplicity null.

Assumptions and the direction of their error. The approximation is exact only under (i) an approximately Gaussian null and (ii) independence among the n attempted candidates. Neither holds exactly. (i) The permutation / `L_fake` nulls are not perfectly Gaussian; the blended Euler-γ form is a first-order correction, not a guarantee. (ii) The candidates a real search tries are typically **positively correlated** (overlapping sign-value assignments). Under a common-variance Gaussian comparison model, positive dependence lowers the expected maximum relative to independence (a Slepian-type comparison; Slepian, 1962), so using the raw attempted-candidate count n in place of a smaller effective independent-trial count places the bar *above* the correlated truth; this does **not** establish conservatism for arbitrary heterogeneous or non-Gaussian dependence — the empirical best-of-100 calibration therefore carries the operational error-control claim. We use the raw deduplicated candidate count; a dependence-adjusted effective count is a recorded, deferred refinement (§9.6) whose correction would only strengthen the nulls.

Calibration and its interval. Because the analytic bar rests on assumptions that do not hold exactly, the error-control claim is carried not by the formula but by the direct empirical calibration (§3.4): a Monte-Carlo of the full gate on a best-of-100 random-map null-generating process (`scripts/gate_null_calibration.py`). The interval on that calibration is the **one-sided exact Clopper–Pearson** bound (the Beta-quantile inversion of the binomial), which needs no normal approximation to the calibration count — for 3 graduations in 500 replicates, a 95% upper bound of $\approx$ 1.54%; the verdict partition of the 500 replicates is in Table 11 (no INCOMPLETE, the search being instrumented). Bailey and López de Prado (2014) supply the selective-inference lineage (§2.4) and the blended expression above; the operational error-control claim nonetheless rests on the empirical calibration, not on the formula.

Cognate-ID baseline (§3.4). The match-rate the deflation is applied to is supplied by the published cognate-ID matcher identified in §2.3 (reproduced in the artifact), so the gate does not depend on any single citation.

Software & data. The artifact. Code, preregistrations, and findings are released as an anonymized artifact on acceptance. Licensed corpora (GORILA-, SigLA-, lineara.xyz-, DĀMOS-derived) are not redistributed; a data-provenance note documents each source's licence and retrieval path.

E Complementary tables

Stratum	**Inscr.**	**Words**	**Sites**	**Role**
admin	290	1,486	14	induction
libation (stone vessels)	99	370	15	induction
other	129	187	40	induction
seal	740	—	—	excluded
emptied by abbrev.-strip	83	—	—	excluded
total	1,341			

Table 2: Morphology corpus strata (Appendix A.1). Seals are structurally excluded as an abbreviation channel; the strata reconcile exactly to the 1,341-inscription corpus total.

Bucketing	$\alpha = 0.01$	$\alpha = 0.05$
strict (stone vessels only)	NULL	NULL
loose (stone objects + inked)	NULL	`i-`, `-te`/`-ti` validate

Table 3: Morphology sensitivity 2×2 (Appendix A.1): genre bucketing $\times$ significance threshold. Only the doubly-relaxed corner validates the H3 nominal affixes `i-` (locative) and `-te`/`-ti` (ablative).

Quantity	**Value**
Linear A confirm rate (16 affixes)	0.250 (4/16; [0.10, 0.50])
`L_fake` bigram-floor rate	0.375 (6/16; [0.18, 0.61])
`i-*301` vs. within-form null	$z \approx 8.0$ (6 environments)
Linear B (DĀMOS) confirm rate	0.562 (9/16)
Linear B within-word-shuffle floor	~0.03
Linear B `L_fake` bigram floor	~0.30 (5 draws)
word-token mean length	1.84 signs (median/mode 1)
word tokens $\leq$2 signs / 1 sign	76.1% / 56.5%
multi-sign *type* mean length	3.07 (Fuls 3.3, 554-type list)
Linear B mean length	3.23 (13,562 wordforms)
DP-unigram segmenter micro-F1	0.436
random-boundary baseline	0.389 (base rate 0.406)
real−random gap, 95% CI	[0.021, 0.099]; P(gap>0) = 0.998
Morfessor	0.156 (over-segments; not usable)

Table 4: Morphology-induction replication quantities (Appendix A.1): confirm rates with Wilson 95% intervals, word-length denominators, and the segmentation positive with its site-clustered bootstrap gap (4,000 draws over 52 sites).

Quantity	**Value**
documents / balance units	32 / 35
integer-level balance	7/28 (25%)
determinable fraction value	J = 1/2 (HT104: $2 \cdot J = 1$)
held-out fraction balance (real)	0.0
permutation null	mean 0.113 (max 0.143); $p = 1.0$
planted positive control	0.9 vs. null 0.22; $p = 0.003$

Table 5: Metrology (Direction D) replication quantities (Appendix B.1). The real corpus is not separated from the permutation null; the planted control separates, so the null is not a machinery bug.

Test	**Classes**	**Acc.**	**Null**	p
consonant class	13	0.080	0.065	0.40
vowel class	5	0.140	0.201	0.86

Table 6: Distributional-phonology per-test results (Appendix B.2): leave-one-out 1-NN vs. a label-permutation null, Šidák-deflated over the two tests. The positive-control power curve fires ($p < .05$) at planted strength ≥ 2 but not at 1.

Leg	**Accuracy**	**Floor**
classical, direct-NN	0.410 [0.18, 0.64]	0.0135
classical, Procrustes	0.185 [0.00, 0.36]	0.0135
I-JEPA (3 seeds; excluded)	0.324 ± 0.025	0.0135
font-swap, direct-NN	$0.367 \to 0.416$	$\gg$ chance
font-swap, Procrustes	$0.156 \to 0.195$	$\gg$ chance
sequence/cognate (DĀMOS)	0.025	0.011
positive control	0.947	—

Table 7: Cross-script sign-image alignment and controls (Appendix B.3). Font-swap rows give same-font $\to$ independent-fonts accuracy on the control's matched anchor set; the sequence/cognate row is the best of five alignment methods over 56 anchors (`clearly_above_chance` = False).

Item	Source	LLM	Baseline
JA-NE = *yn* "wine"	Gordon	14/40	0/40
A-SA-SA-RA-ME	Best (1981)	2/40	0/40
KA-RO-PA	Gordon	0/40	0/40
SU-PA-RA	Gordon	0/40	0/40
KU-RO = *kull* (downwt.)	Gordon	8/40	—
KU-RO = "total" (excl.)	gen. knowledge	7/40	—

Table 8: LLM-ablation per-item reproduction over 40 seeds, qwen2.5:72b vs. the model-free edit-distance baseline (Appendix C.1). The two KU-RO routes overlap on one seed; three single-word items are among Gordon's five West-Semitic equations, while A-SA-SA-RA-ME is Best's.

Sign set	Signs	Valued (of 40)
known AB (indexed)	20	$\sim$20
not-indexed *-series	11	$\sim$1 ($\sim$97% abstention)

Table 9: L_not_indexed abstention asymmetry (Appendix C.2). The apparent consistency gap 0.601 (permutation $p = 0.0005$) is coverage-confounded; the matched-n check returned no_power (0 of 11 not-indexed signs reached $n \geq 10$).

Denominator / channel	Value
token-frequent syllabograms	97
full syllabary estimate	$\sim$150
working repertoire (V; conflates syll./subscr./logograms)	259
incl. ideograms (GORILA: 390)	$\sim$350
sign occurrences, all types	$\sim$7,400
parsed syllable-signs (unicity input)	$N \approx 5{,}792$
corpus envelope	$\sim$2,500 finds; $\sim$1,400 readable
findplaces (logos-normalized)	56 ($\rightarrow$ 52 sites)
readable inscriptions	1,341
distinct word-forms	539
unicity distance over (V, P) space	$U \approx$ 204–415 signs

Table 10: Corpus denominators, occurrence channels, and the unicity range (Appendix C.4). The $\sim$7,400 all-type occurrence channel never enters the unicity arithmetic, which consumes only the parsed syllable-signs.

Calibration quantity	Value
S_morph FPR (within-form shuffle)	3.0% (6/200; [1.4, 6.4]%)
nominal one-sided $z \geq 2$ rate	$\approx$2.3%
S_morph power (planted affix)	100% (50/50; saturated)
gate false-graduation (best-of-100 null)	0.6% (3/500; CP $\leq$ 1.54%)
gate verdict partition (500 replicates)	3 GRAD / 10 NULL / 487 REJ

Table 11: Detector and gate calibration quantities (Appendices A.2 and D): Wilson interval for the S_morph false-positive rate; one-sided exact Clopper–Pearson 95% upper bound for the gate's false-graduation rate under the tested best-of-100 random-map null.

Quantity	Value
literature-index census	63 claims / 62 entries (55 single)
root-template TV distance	$\sim$0.84 $\rightarrow$ $\sim$0.33 ($\sim$0.23 standalone)
ablation scale	40 seeds $\times$ 40 held-out forms
model-free lexicon	7,682 Hebrew skeletons
LLM errors	0 (40/40 seeds)
value-vocabulary overlap	10 of 134
first-pass contamination rate	1.000 (33/33 defined; artifact)
gate-2 rate	mean 0.646 (confounded; not reported)

Table 12: Literature-index, L_fake-canary, and LLM-ablation quantities (Appendices C.1 and C.5). The TV distance is measured end-to-end on the Ugaritic$\leftrightarrow$Hebrew gold; the first-pass contamination rate is forced to 1.0 by construction and reported as an artifact.